



SCAN TO PLAY WITH THIS SLIDE

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1 of 20

SPACE SCIENCE & ASTRONOMY FOR OUTREACH & EDUCATION

Visualizing the Transient Sky: Real-Time All-Sky Imaging and Open Data in the TART Ecosystem

T Molteno, B Hugo, S Mirza, [MYA Khondoker](#), SR Diya, SH Abrar, SH
Muzaffar, MFH Sami, T Rahman, MH Masum, KMB Asad

PSSARC 2026 • University of San Carlos • Cebu, Philippines • CASSA / Independent University, Bangladesh



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The TART Telescope





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CASSA Workshop 2 • Building & Learning TART





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Overview of the talk

Interferometry: instead of one huge dish, link many small antennas — compare the waves each pair catches, and the maths turns them into a single giant telescope.

01

Interferometry

An interferometer - fringe

Antennas and u-v plane

Visibility Vectors

Visibility Complex Conjugates

All Pairs in the Array - fringe

Software Pipeline for Producing an Image

02

Software Pipeline

Installation



Downloading Visibilities



Measurement Set



Antenna and UV plane



Amplitude Calibration



Phase Calibration



Apply Calibration



CLEAN

03

The Hardware

Presented in the later talk



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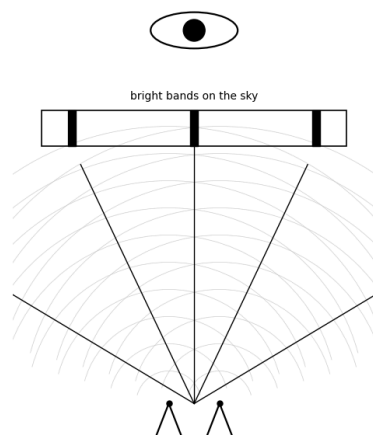
— or visit —

cassa-talks-tart-flame.vercel.app

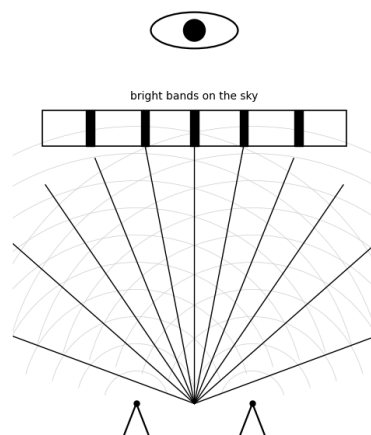
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An interferometer - fringe

close antennas → bold, wide-spaced

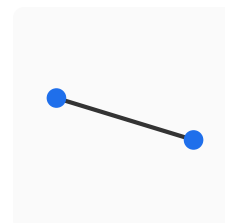


far antennas → fine, tightly-packed



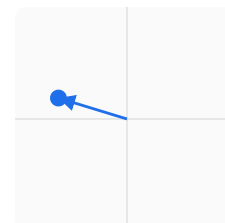
Wavefronts arriving at two antennas: where they add, a bright band; where they cancel, a dark one. Close pair → bold bands. Far pair → fine bands.

① the two antennas



1.20 m apart

② one dot in the uv-map



6.3 wavelengths from centre

③ one fringe on the sky



stripe spacing ≈ 0.16 (finer as the dot moves out)

Distance: **1.20 m**



Orientation: **163°**



Slide **Separation** → the dot moves out and the stripes get finer. Slide **Orientation** → the dot circles the centre and the stripes turn with it, always at a right angle to the antenna pair.

↑ These two sliders are live — scan the QR and drag them yourself.



SCAN TO PLAY WITH THIS SLIDE

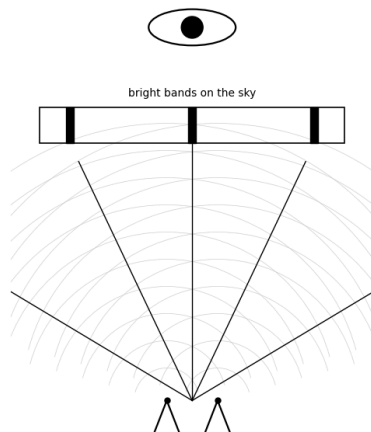
— or visit —

cassa-talks-tart-flame.vercel.app

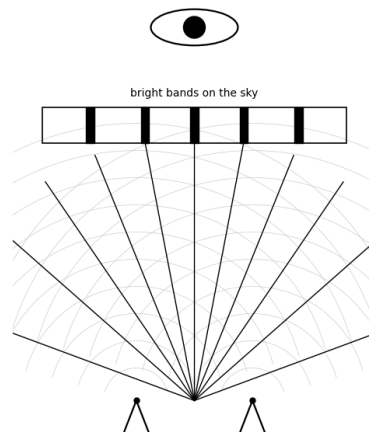
5 of 20

An interferometer - fringe

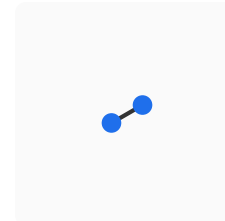
close antennas → bold, wide-spaced



far antennas → fine, tightly-packed

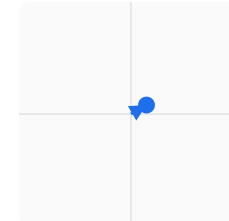


① the two antennas



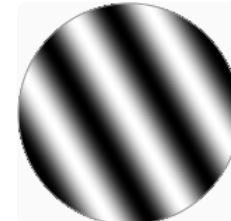
0.30 m apart

② one dot in the uv-map



1.6 wavelengths from centre

③ one fringe on the sky



stripe spacing ≈ 0.63 (finer as the dot moves out)

Distance: **0.30 m**



Orientation: **30°**



Slide **Separation** → the dot moves out and the stripes get finer. Slide

Orientation → the dot circles the centre and the stripes turn with it, always at a right angle to the antenna pair.

Close pair, 0.30 m apart — bold, widely spaced bands.



SCAN TO PLAY WITH THIS SLIDE

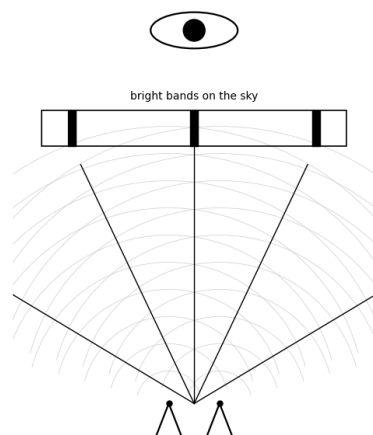
— or visit —

cassa-talks-tart-flame.vercel.app

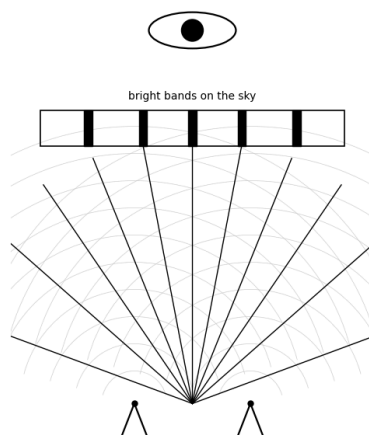
5 of 20

An interferometer - fringe

close antennas → bold, wide-spaced

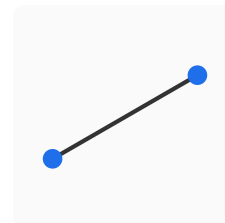


far antennas → fine, tightly-packed



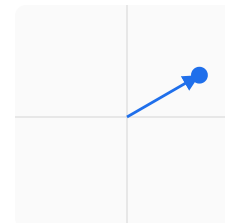
Same orientation, moved apart to 1.40 m — the bands get finer.

① the two antennas



1.40 m apart

② one dot in the uv-map



7.4 wavelengths from centre

③ one fringe on the sky



stripe spacing ≈ 0.14 (finer as the dot moves out)

Distance: **1.40 m**

Orientation: **30°**

Slide **Separation** → the dot moves out and the stripes get finer. Slide **Orientation** → the dot circles the centre and the stripes turn with it, always at a right angle to the antenna pair.



SCAN TO PLAY WITH THIS SLIDE

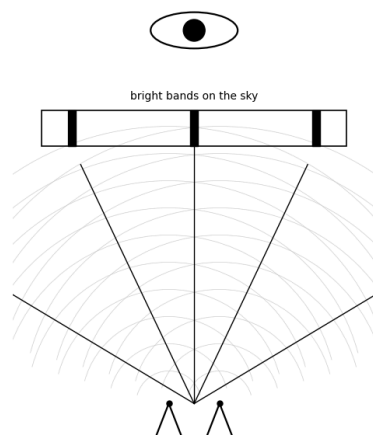
– or visit –

cassa-talks-tart-flame.vercel.app

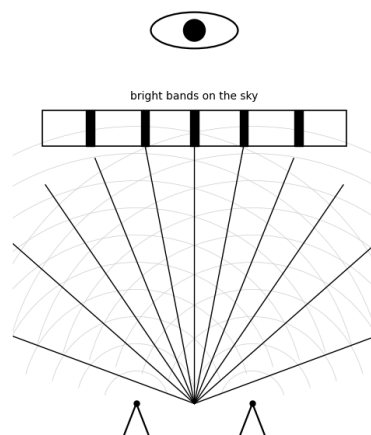
5 of 20

An interferometer - fringe

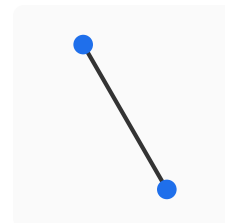
close antennas → bold, wide-spaced



far antennas → fine, tightly-packed

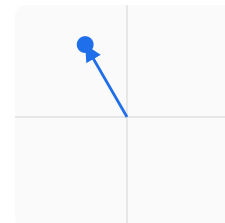


① the two antennas



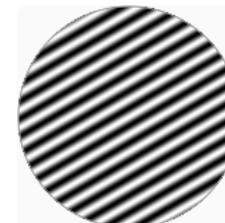
1.40 m apart

② one dot in the uv-map



7.4 wavelengths from centre

③ one fringe on the sky



stripe spacing ≈ 0.14 (finer as the dot moves out)

Distance: **1.40 m**

Orientation: **120°**

Slide **Separation** → the dot moves out and the stripes get finer. Slide

Orientation → the dot circles the centre and the stripes turn with it, always at a right angle to the antenna pair.

Same 1.40 m separation, pair rotated to 120° — the bands turn with it.



SCAN TO PLAY WITH THIS SLIDE

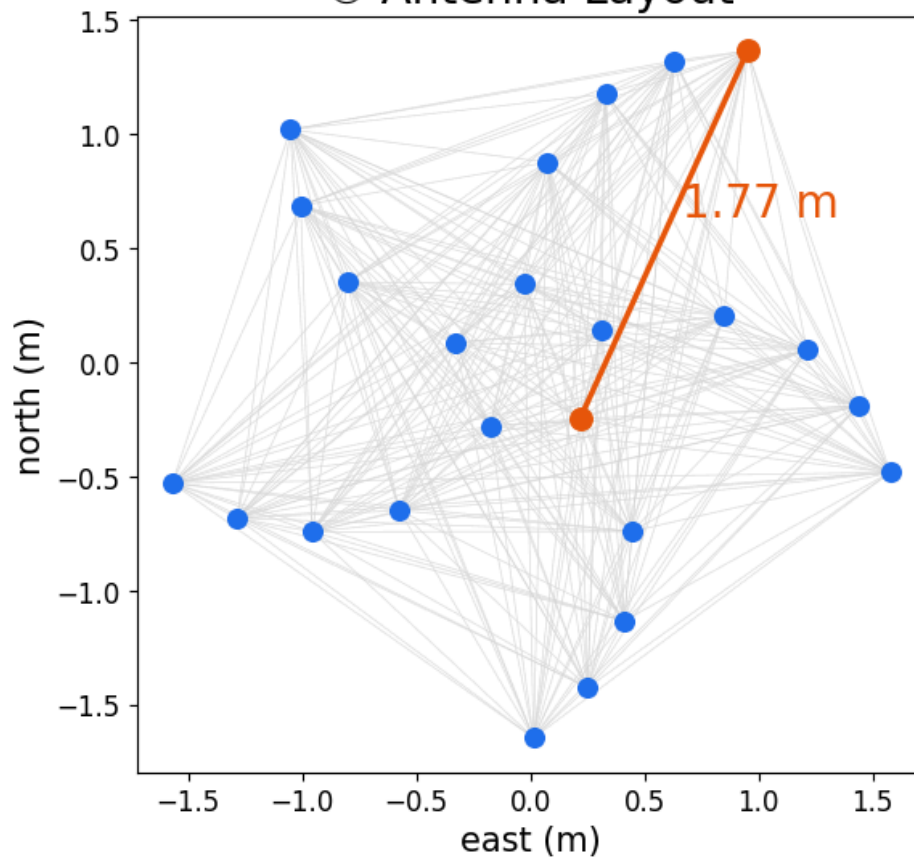
— or visit —

cassa-talks-tart-flame.vercel.app

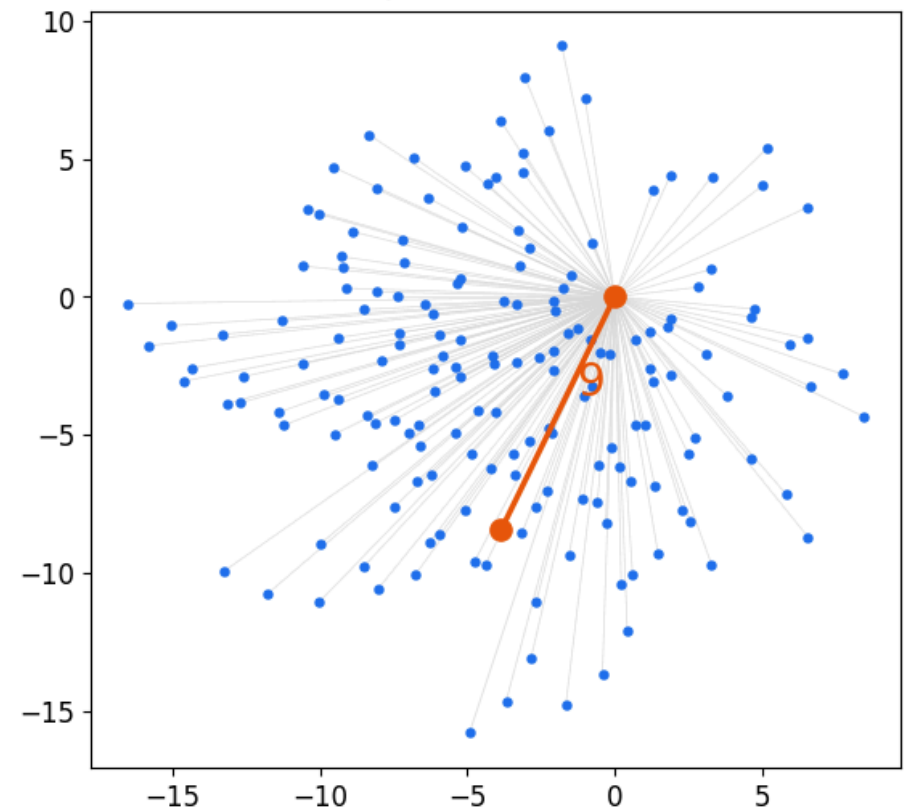
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Antennas and u-v plane

① Antenna Layout



② UV Plane





SCAN TO PLAY WITH THIS SLIDE

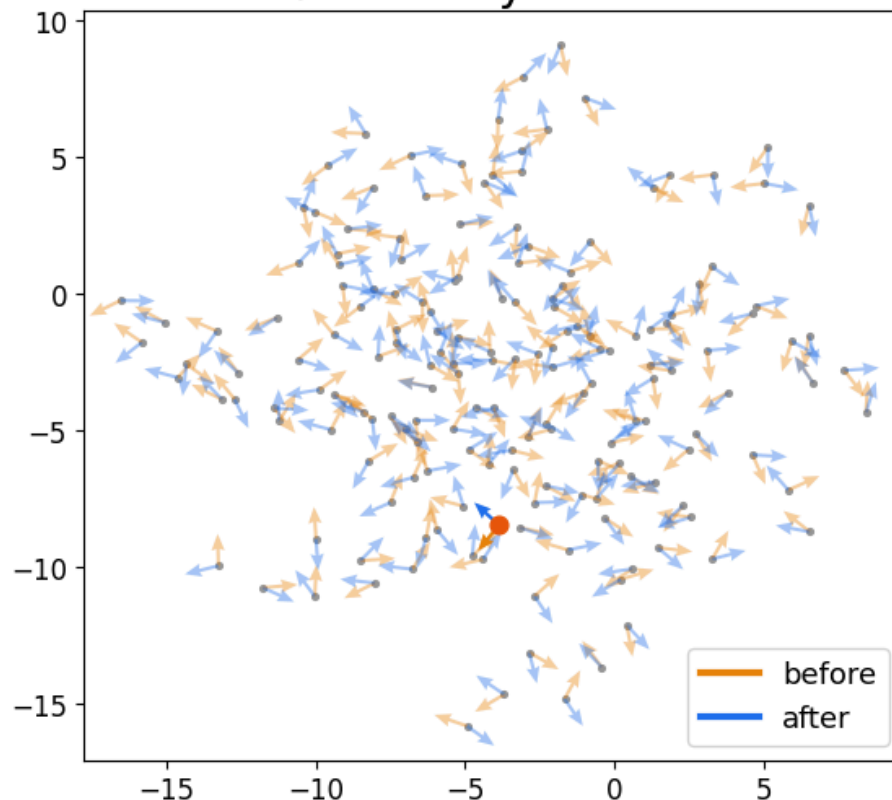
– or visit –

cassa-talks-tart-flame.vercel.app

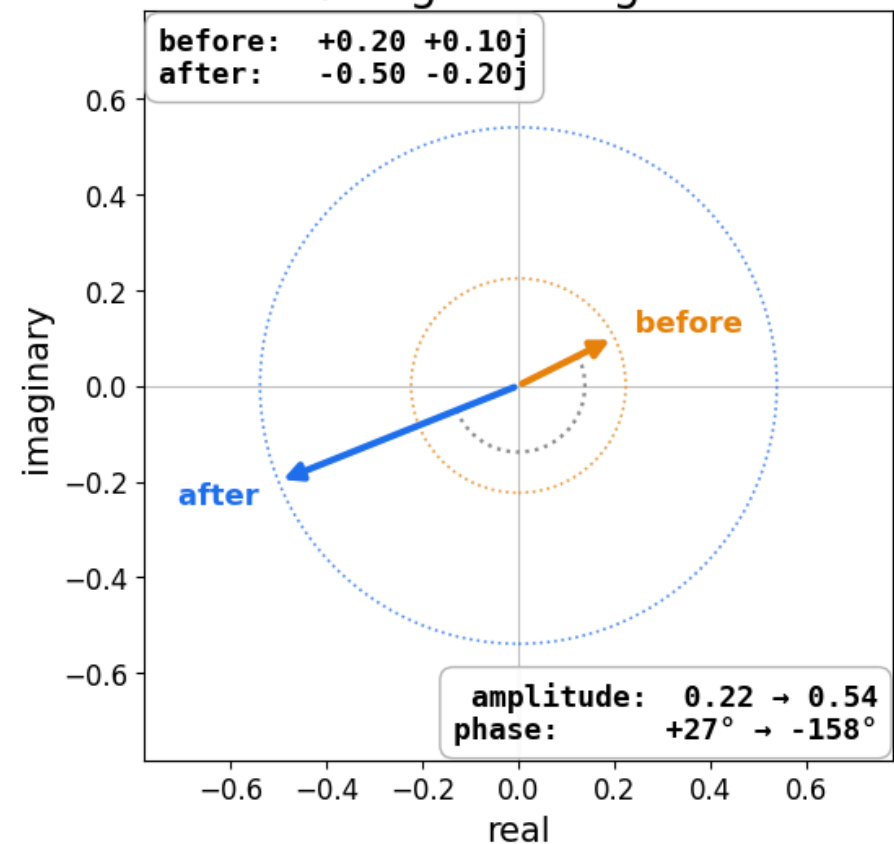
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Visibility Vectors

③ Visibility Vectors



④ Argand Diagram





SCAN TO PLAY WITH THIS SLIDE

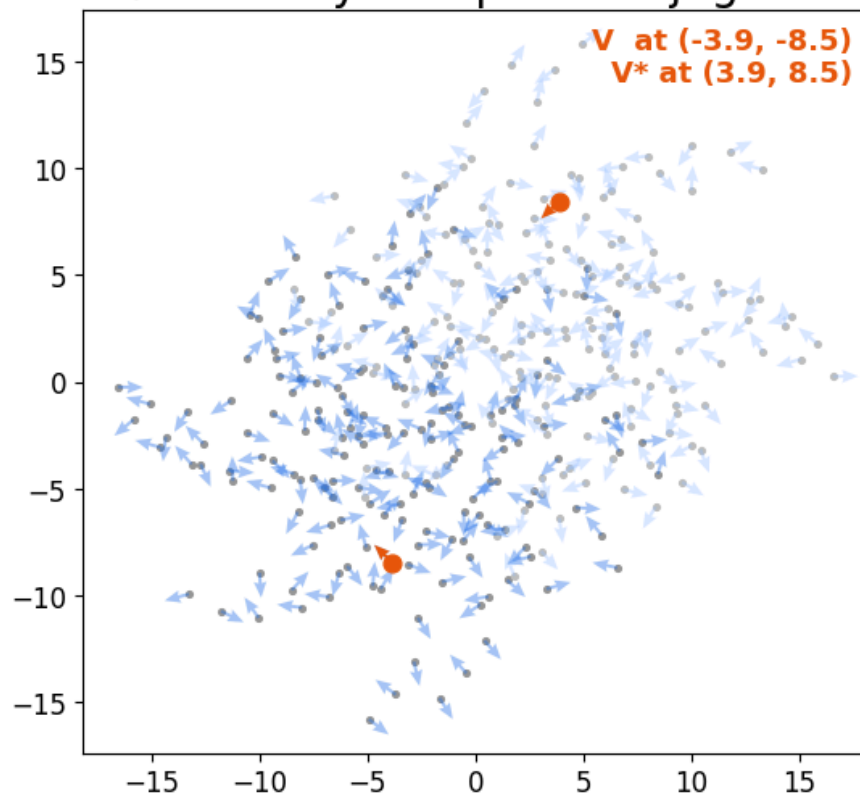
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cassa-talks-tart-flame.vercel.app

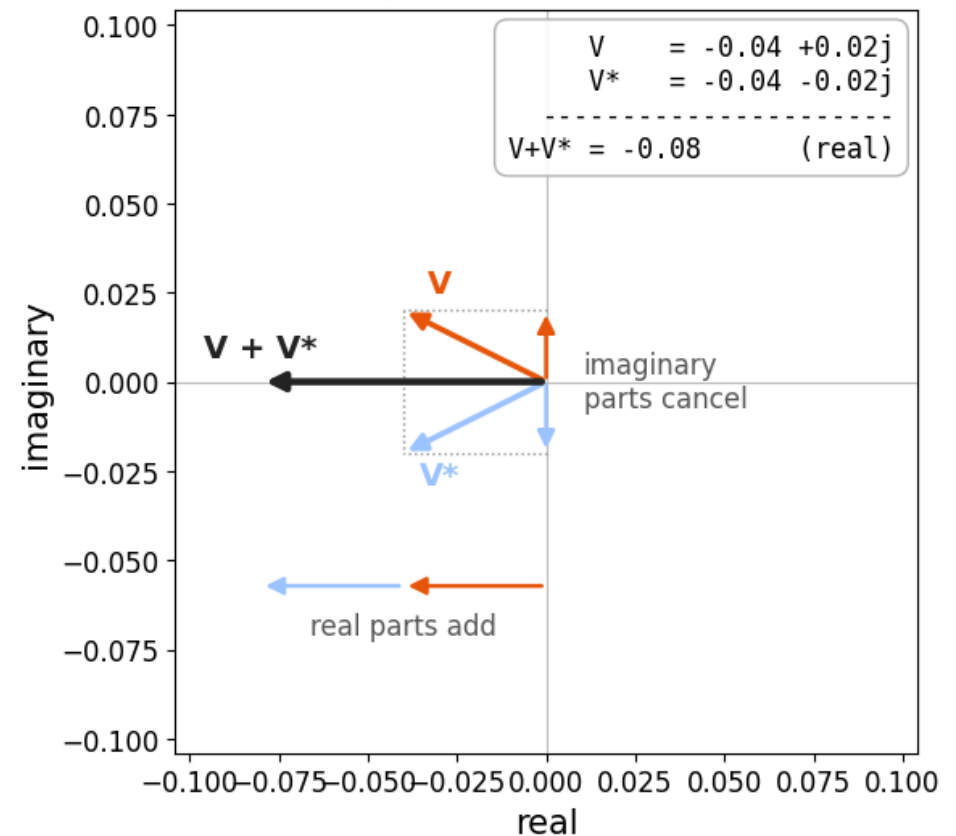
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Visibility Complex Conjugates

⑤ Visibility Complex Conjugates



⑥ $V + V^*$ is real





SCAN TO PLAY WITH THIS SLIDE

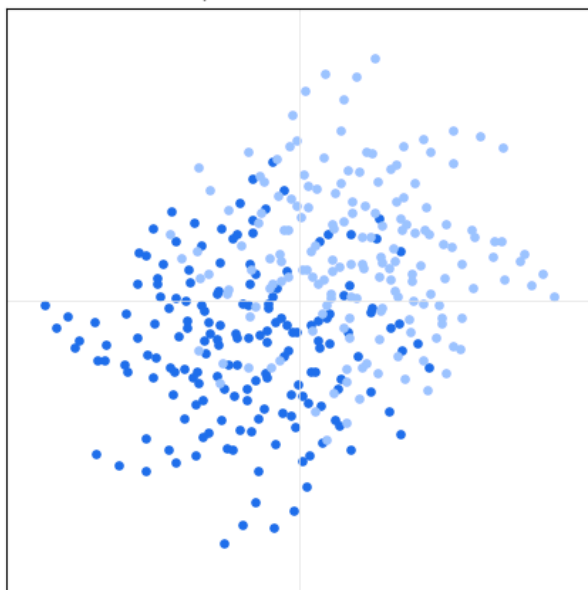
– or visit –

cassa-talks-tart-flame.vercel.app

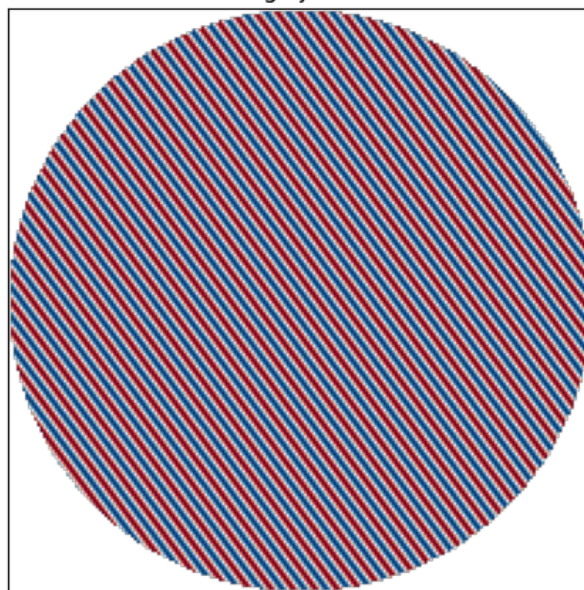
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All Pairs in the Array - fringe

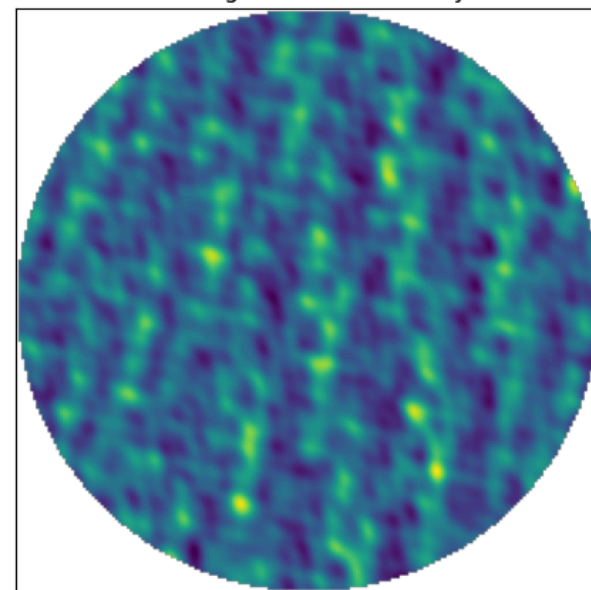
171/171 baselines used



the fringe just added



all fringes added → the sky



drag → add baselines one by one



SCAN TO PLAY WITH THIS SLIDE

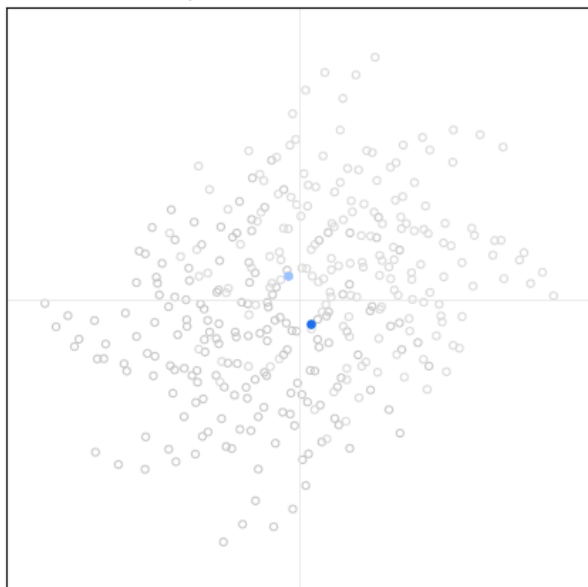
– or visit –

cassa-talks-tart-flame.vercel.app

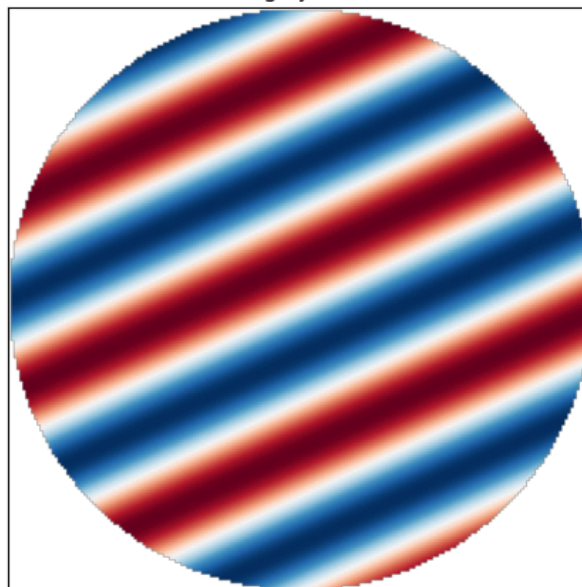
9 of 20

All Pairs in the Array - fringe

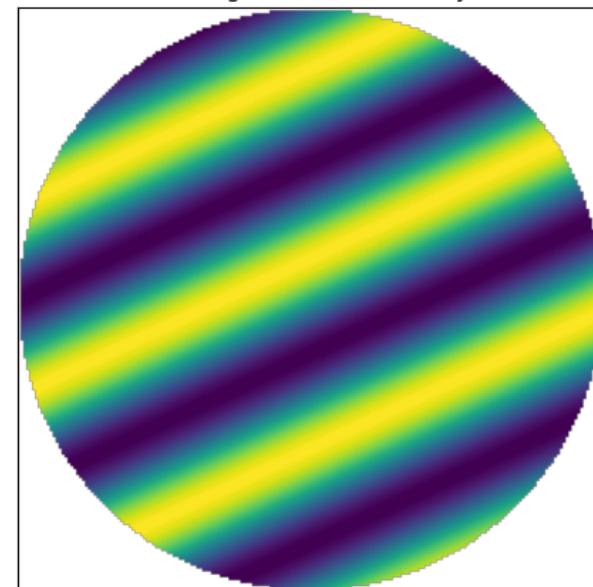
1/171 baselines used



the fringe just added



all fringes added → the sky



1 of 34 baselines added



SCAN TO PLAY WITH THIS SLIDE

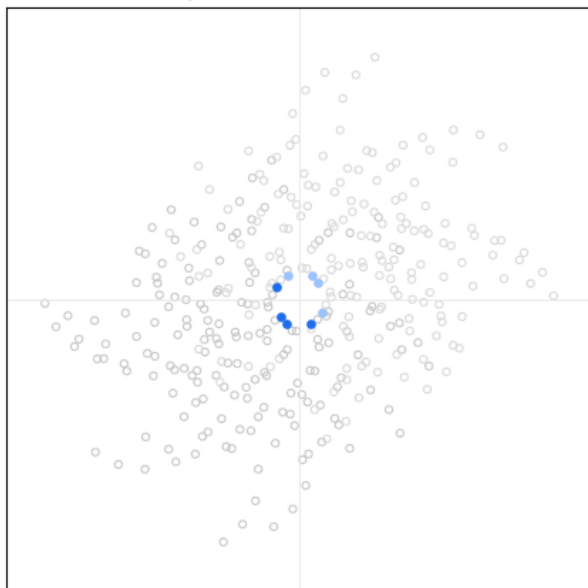
– or visit –

cassa-talks-tart-flame.vercel.app

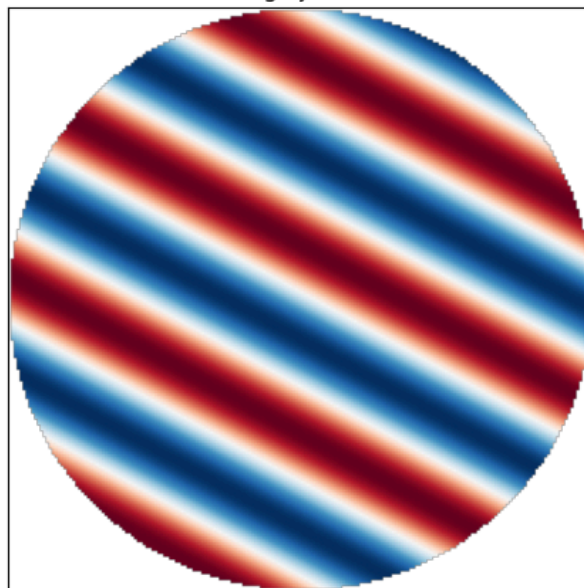
9 of 20

All Pairs in the Array - fringe

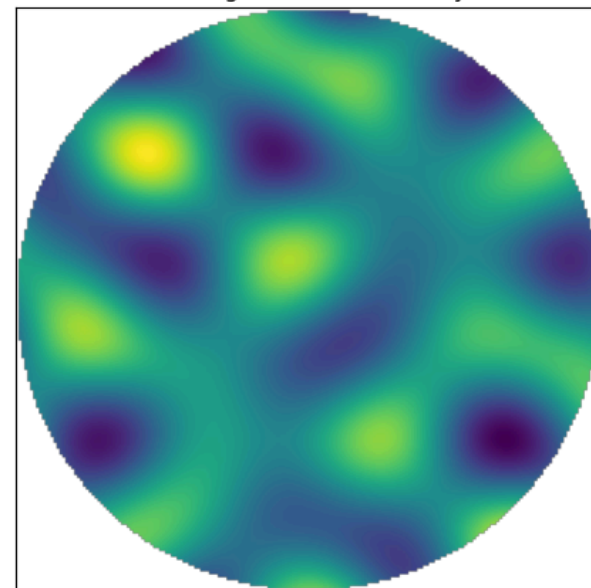
4/171 baselines used



the fringe just added



all fringes added → the sky



4 of 34 baselines added



SCAN TO PLAY WITH THIS SLIDE

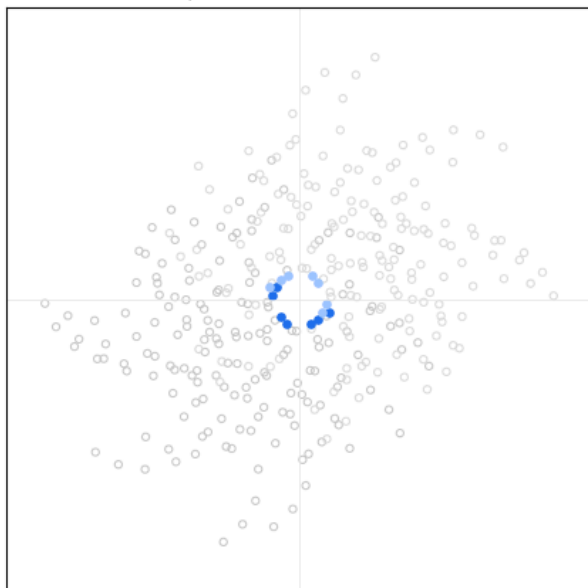
– or visit –

cassa-talks-tart-flame.vercel.app

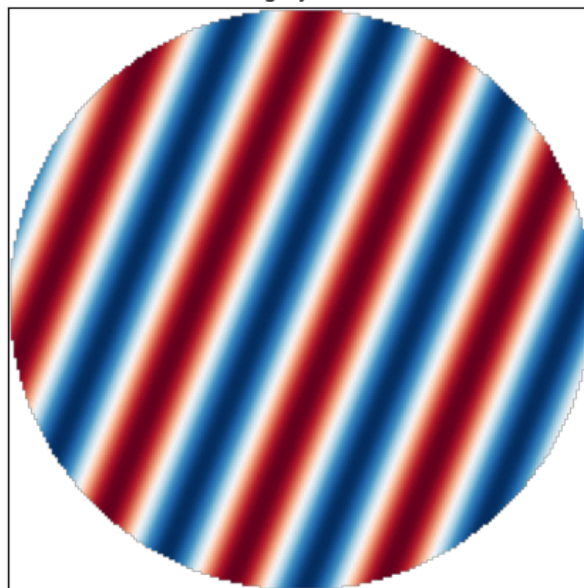
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All Pairs in the Array - fringe

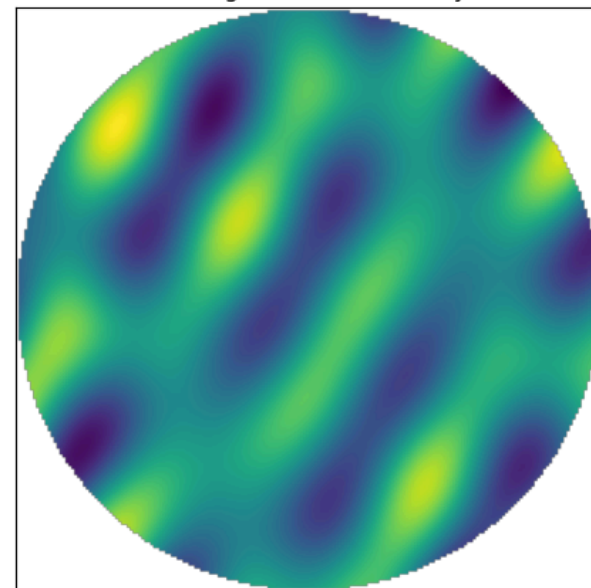
7/171 baselines used



the fringe just added



all fringes added → the sky



7 of 34 baselines added



SCAN TO PLAY WITH THIS SLIDE

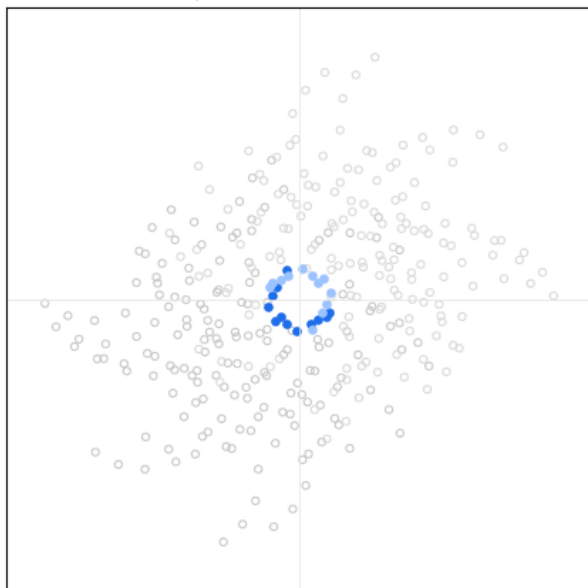
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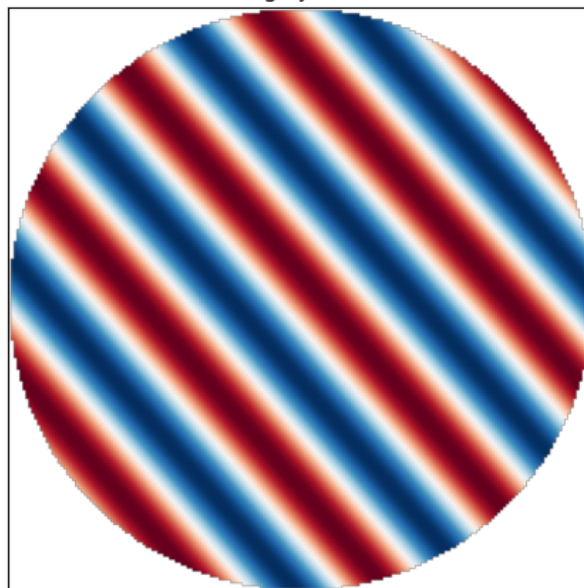
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All Pairs in the Array - fringe

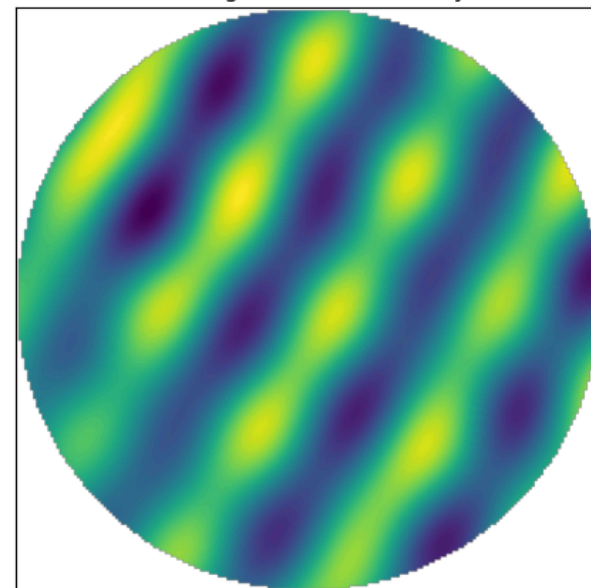
12/171 baselines used



the fringe just added



all fringes added → the sky



11 of 34 baselines added



SCAN TO PLAY WITH THIS SLIDE

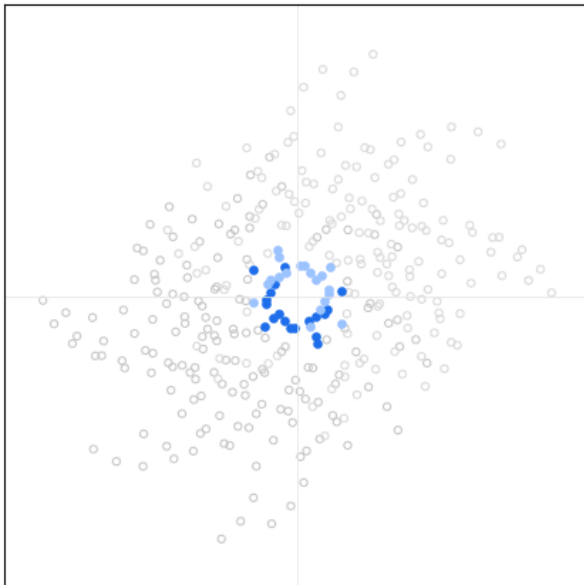
– or visit –

cassa-talks-tart-flame.vercel.app

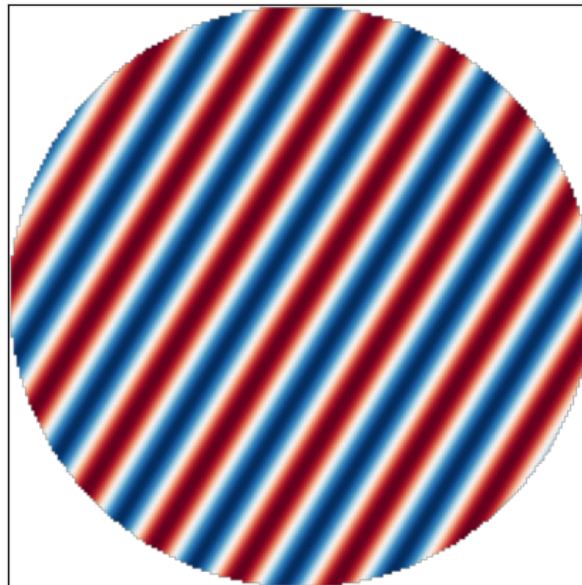
9 of 20

All Pairs in the Array - fringe

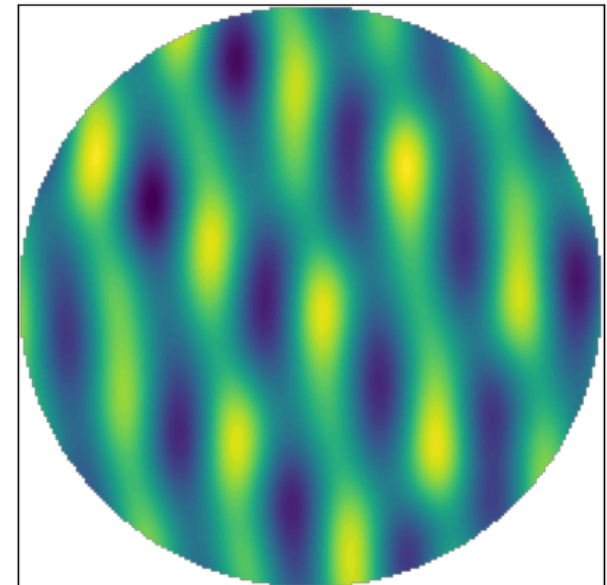
19/171 baselines used



the fringe just added



all fringes added → the sky



15 of 34 baselines added



SCAN TO PLAY WITH THIS SLIDE

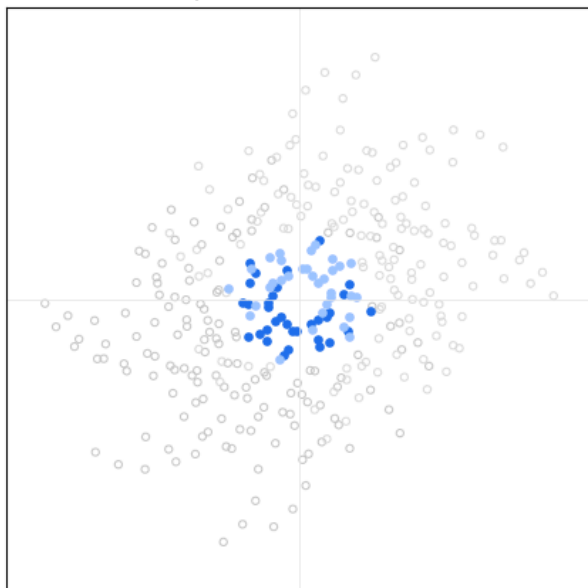
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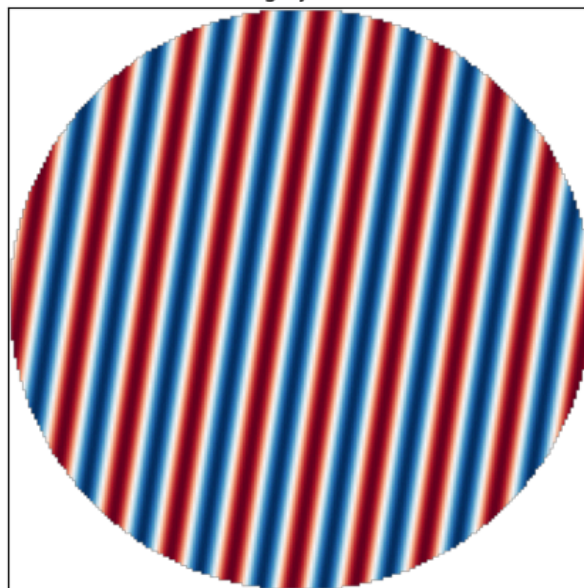
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All Pairs in the Array - fringe

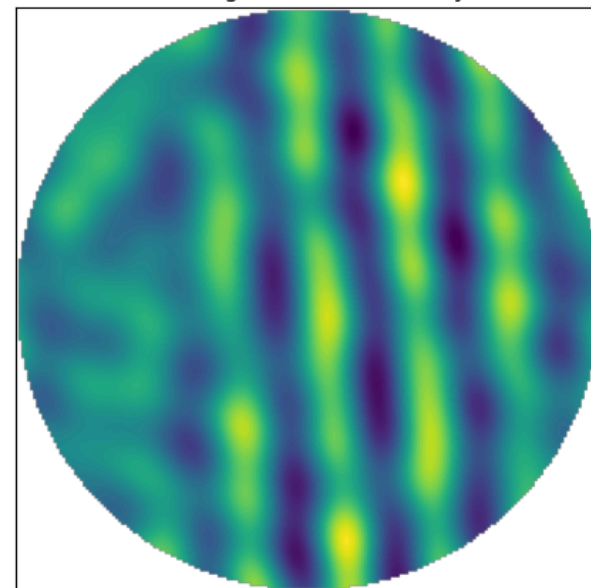
33/171 baselines used



the fringe just added



all fringes added → the sky



20 of 34 baselines added



SCAN TO PLAY WITH THIS SLIDE

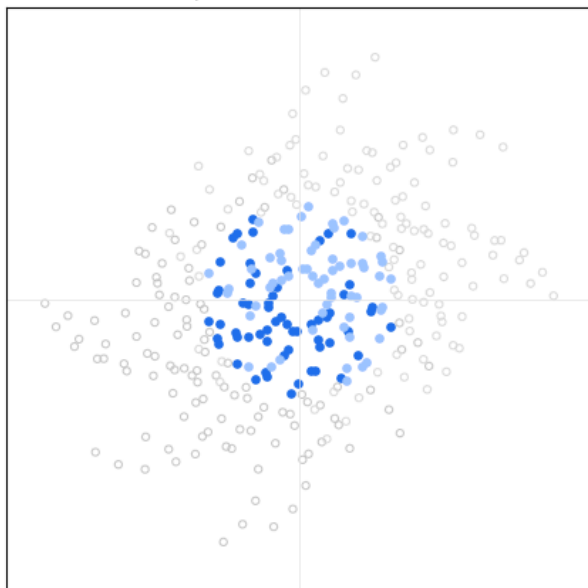
– or visit –

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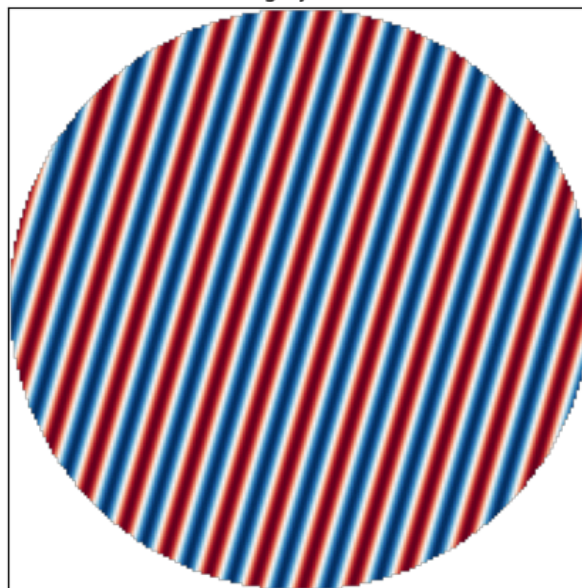
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All Pairs in the Array - fringe

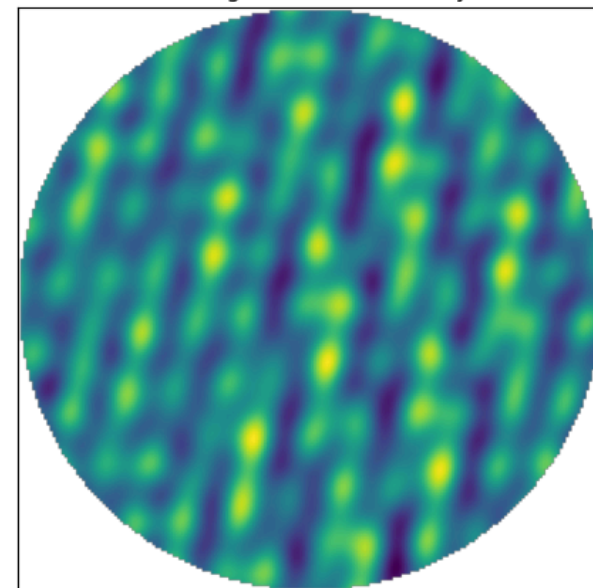
60/171 baselines used



the fringe just added



all fringes added → the sky



25 of 34 baselines added



SCAN TO PLAY WITH THIS SLIDE

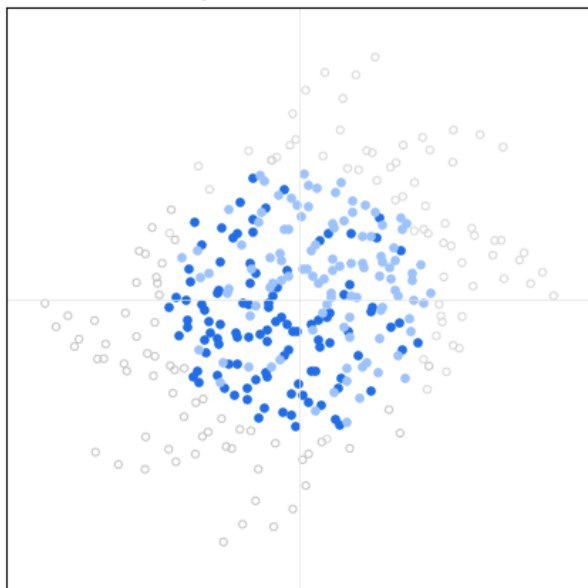
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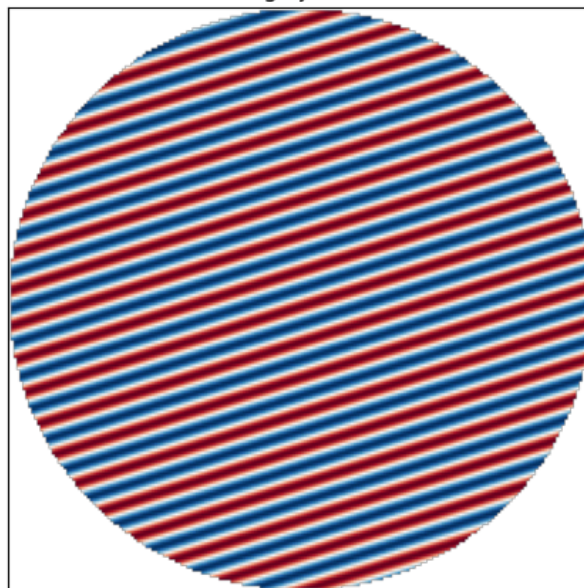
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All Pairs in the Array - fringe

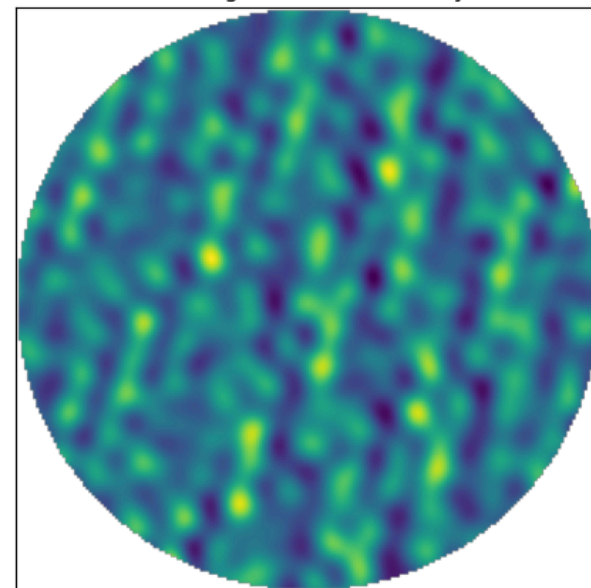
107/171 baselines used



the fringe just added



all fringes added → the sky



30 of 34 baselines added



SCAN TO PLAY WITH THIS SLIDE

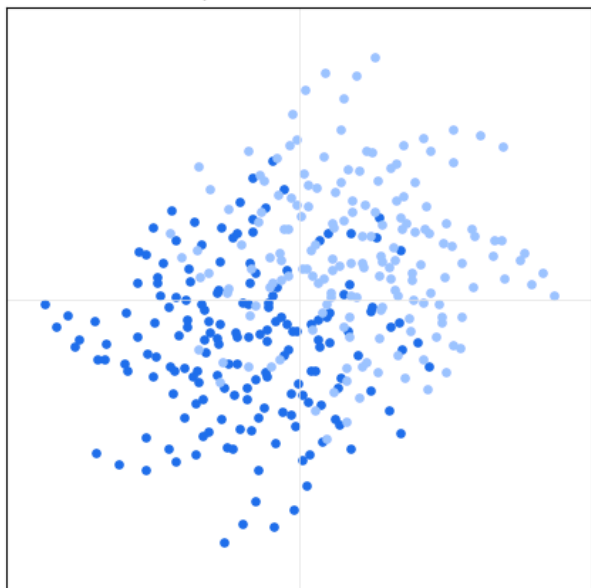
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cassa-talks-tart-flame.vercel.app

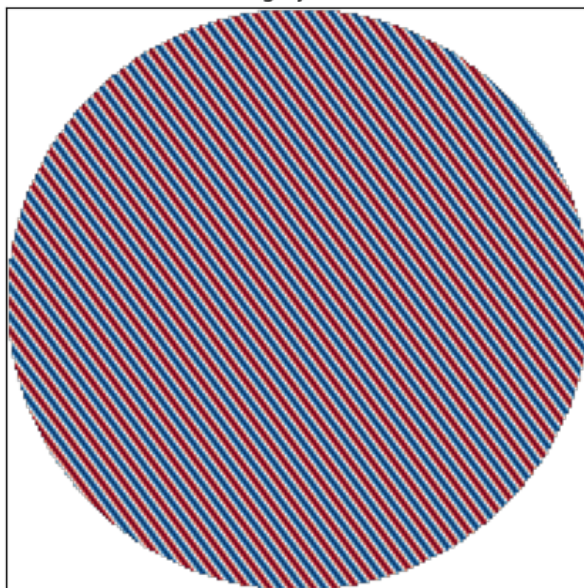
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All Pairs in the Array - fringe

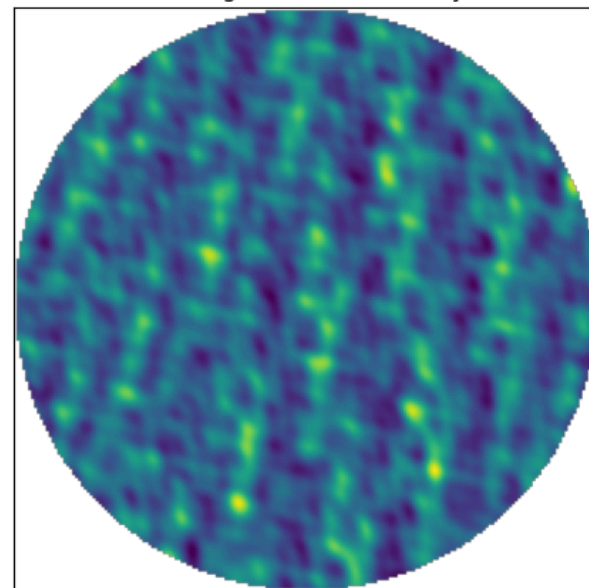
171/171 baselines used



the fringe just added



all fringes added → the sky



34 of 34 baselines added



SCAN TO PLAY WITH THIS SLIDE

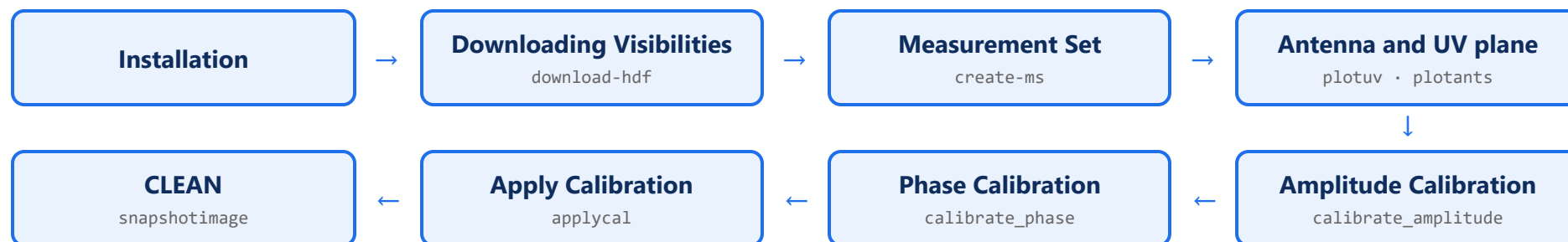
– or visit –

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SETUP

Software Pipeline for Producing an Image



TART MAP & DOCUMENTATION
<https://tart.elec.ac.nz/>

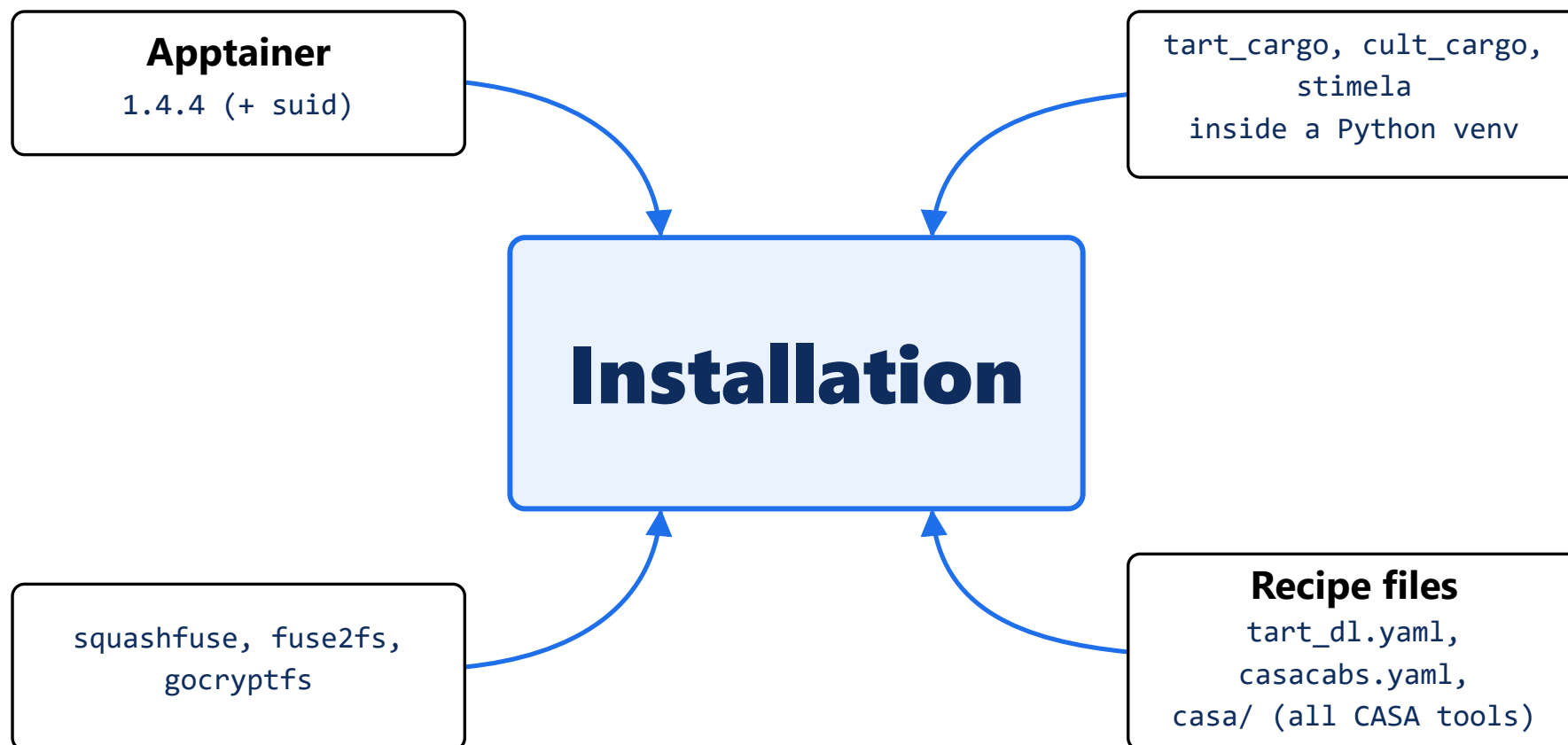
STIMELA RECIPES & DOCUMENTATION
`git clone https://github.com/gwynbleidrivia/tart-stimela-run`



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SETUP





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STAGE 1 • ACQUIRE

Downloading Visibilities

download-hdf

Pulls a bunch of snapshots
off the telescope.

Explore the raw data

<https://api.elec.ac.nz/tart/bd-iub>

[/api/v1/info](#)

[/api/v1/mode/current](#)

[/api/v1/calibration/gain](#)

[/api/v1/imaging/vis](#)

TART MAP & DOCUMENTATION
<https://tart.elec.ac.nz/>

STIMELA RECIPES & DOCUMENTATION
`git clone https://github.com/gwynbleidrivia/tart-stimela-run`



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STAGE 2 • BUILD

Measurement Set

create-ms

ANT1	ANT2	U (λ)	V (λ)	DATA	phase°	MODEL
0	1	0.54	2.80	0.383	65.7	8.602
0	2	1.88	4.37	0.077	-125.4	11.071
0	3	3.47	5.13	0.158	42.4	8.544
0	4	5.15	5.37	0.101	118.1	6.074
0	5	1.77	-1.07	0.212	-52.4	10.236
0	6	4.61	-0.73	0.190	-145.6	9.615
0	7	6.52	-1.52	0.125	-110.3	0.992
0	8	7.71	-2.81	0.126	-42.5	1.256
0	9	8.43	-4.34	0.173	90.3	14.141
0	10	1.29	-3.10	0.110	-46.2	9.976

TART MAP & DOCUMENTATION
<https://tart.elec.ac.nz/>

STIMELA RECIPES & DOCUMENTATION
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SCAN TO PLAY WITH THIS SLIDE

– or visit –

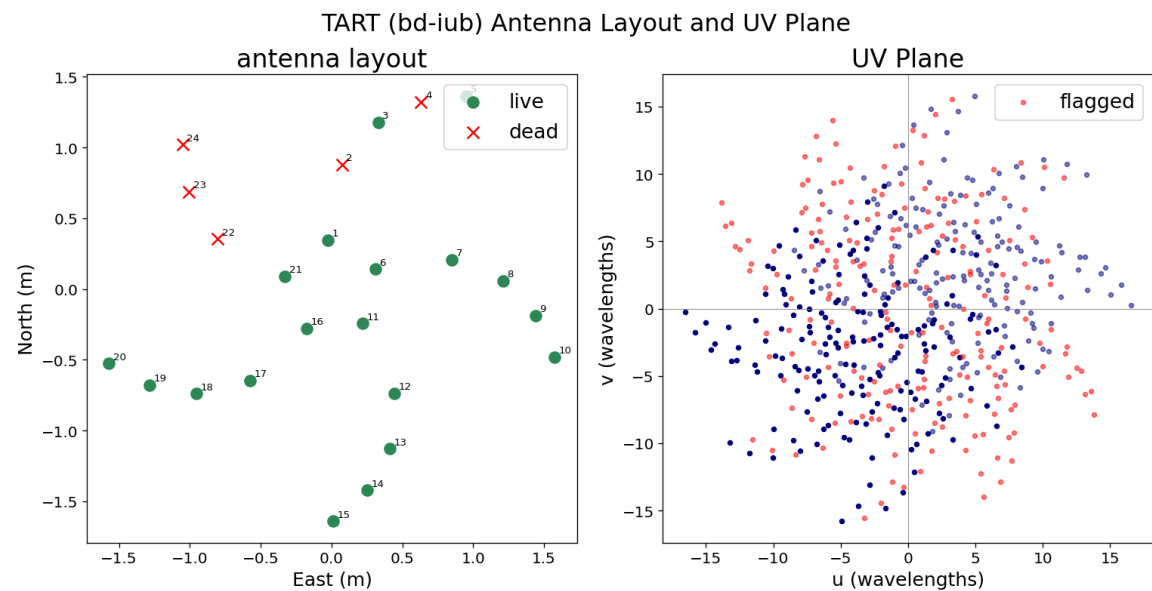
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STAGE 3 • INSPECT

Antenna and UV plane

plotuv
plotants



TART MAP & DOCUMENTATION
<https://tart.elec.ac.nz/>

STIMELA RECIPES & DOCUMENTATION
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SCAN TO PLAY WITH THIS SLIDE

– or visit –

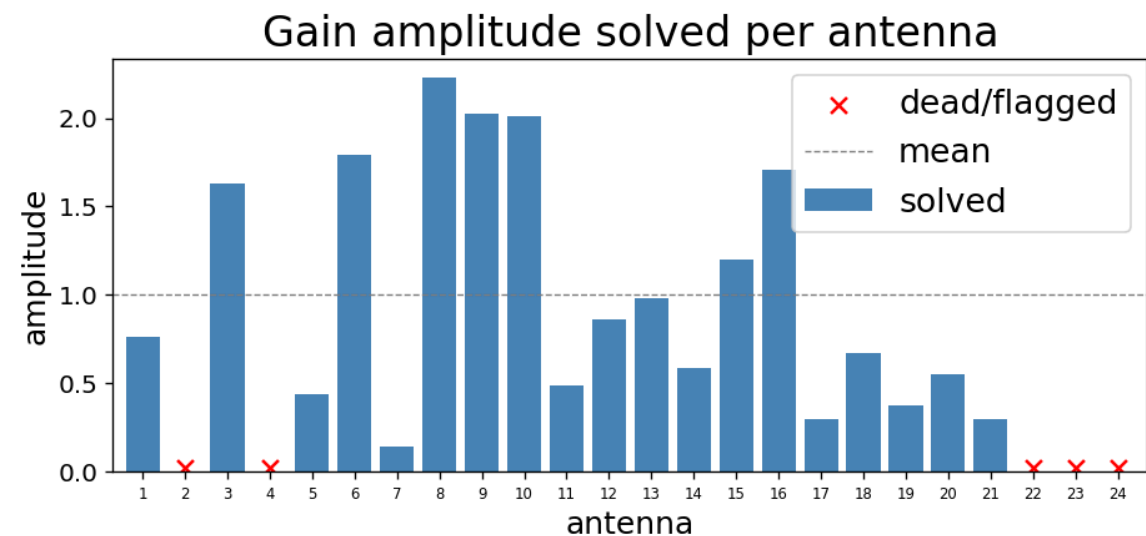
cassa-talks-tart-flame.vercel.app

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STAGE 4 • CALIBRATE

Amplitude Calibration

calibrate_amplitude



TART MAP & DOCUMENTATION
<https://tart.elec.ac.nz/>

STIMELA RECIPES & DOCUMENTATION
`git clone https://github.com/gwynbleidrivia/tart-stimela-run`



SCAN TO PLAY WITH THIS SLIDE

– or visit –

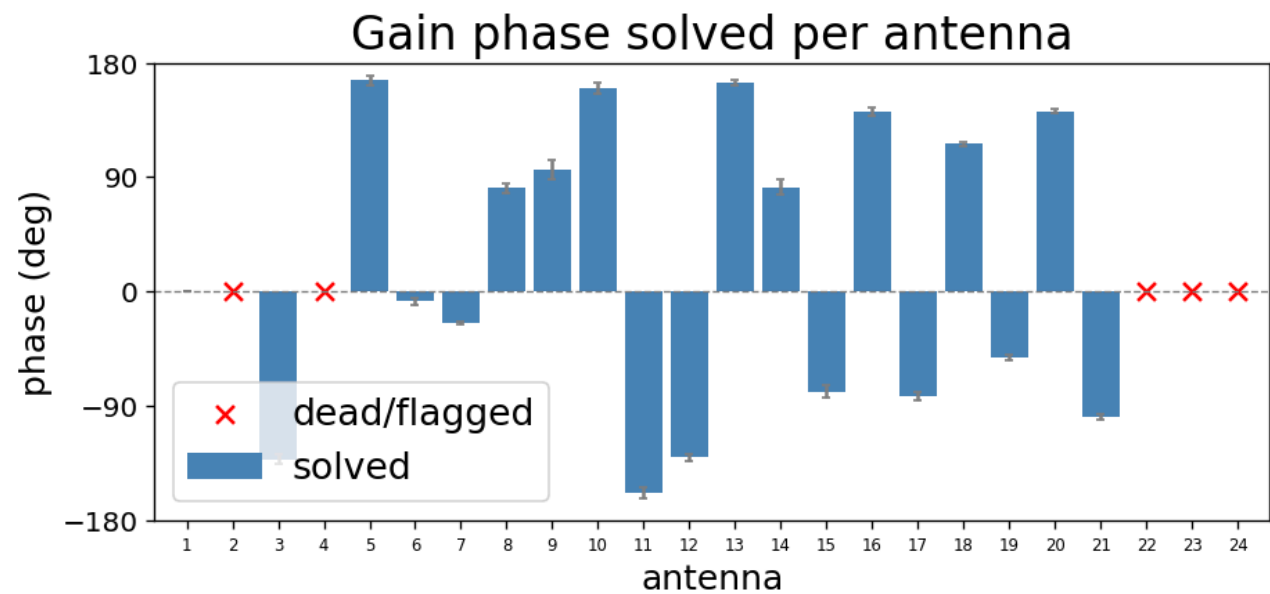
cassa-talks-tart-flame.vercel.app

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STAGE 5 • CALIBRATE

Phase Calibration

calibrate_phase





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– or visit –

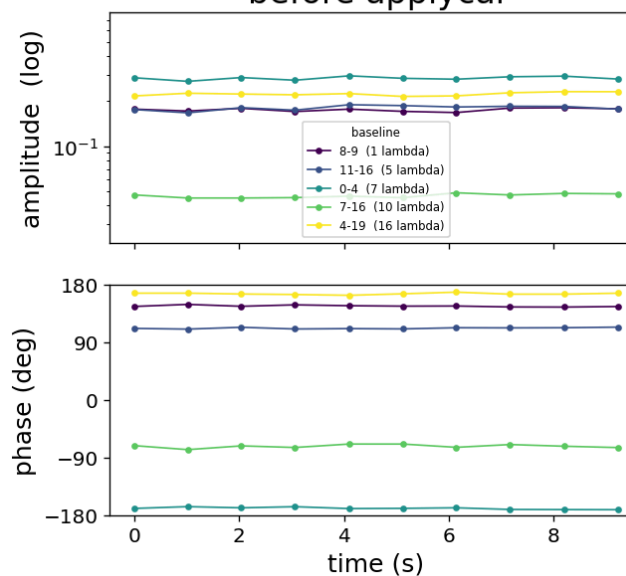
cassa-talks-tart-flame.vercel.app

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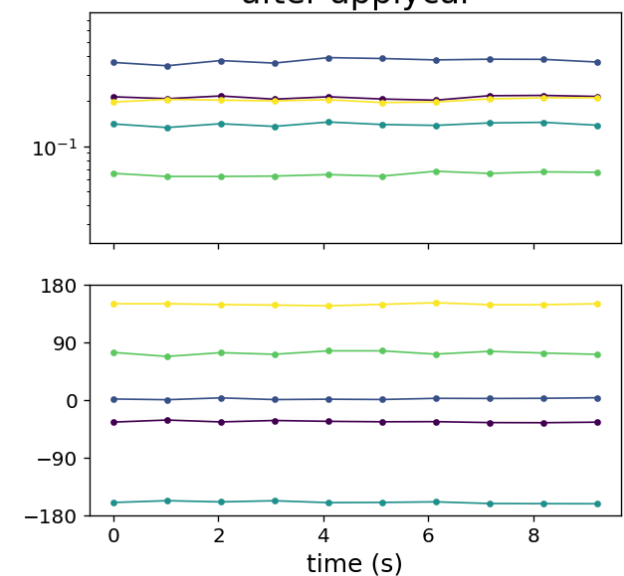
STAGE 6 • APPLY

Visibilities vs Time

before applycal



after applycal



Apply Calibration

applycal



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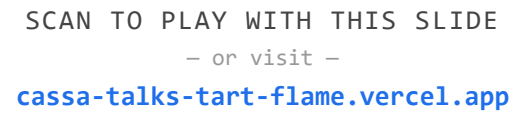
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STAGE 6 • APPLY

Apply Calibration

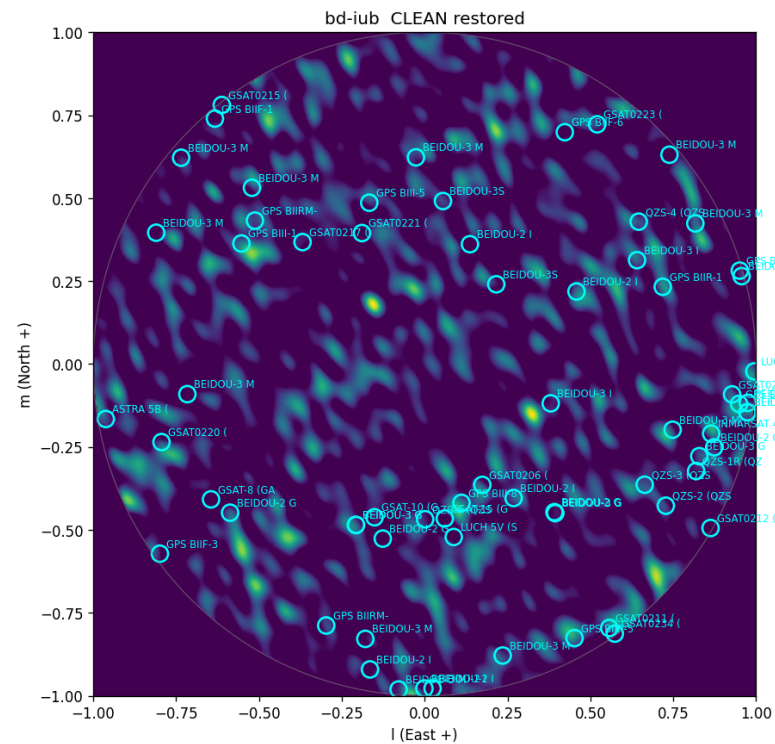
applycal

ANT1	ANT2	U (λ)	V (λ)	DATA	phase°	MODEL	CORRECTED	corr. phase°
0	1	0.54	2.80	0.383	65.7	8.602	0.000	0.0
0	2	1.88	4.37	0.077	-125.4	11.071	0.062	104.8
0	3	3.47	5.13	0.158	42.4	8.544	0.000	0.0
0	4	5.15	5.37	0.101	118.1	6.074	0.304	-73.4
0	5	1.77	-1.07	0.212	-52.4	10.236	0.156	-51.8
0	6	4.61	-0.73	0.190	-145.6	9.615	1.808	-166.4
0	7	6.52	-1.52	0.125	-110.3	0.992	0.074	-27.9
0	8	7.71	-2.81	0.126	-42.5	1.256	0.082	50.4
0	9	8.43	-4.34	0.173	90.3	14.141	0.113	-96.8
0	10	1.29	-3.10	0.110	-46.2	9.976	0.299	171.5



STAGE 7 • IMAGE

CLEAN
snapshotimage





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Under One Sky

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